Design Assignment #8:
DAMNED Assembly

In this module you will assemble the DAMNED project and install code including Python libraries. If you have some experience building things and working with CircuitPython this should be a relatively quick week compared to some of the other design assignments. If not, this design assignment will walk you through some of the necessary steps and the Perusall readings will give you needed background information.

Next week you'll start to get more into CircuitPython coding, connecting some sensors to the interfaces that are built into the project and starting to learn how to collect and analyze data.

Objectives

- Develop competency in assembling a device using hand tools and securing the parts of an enclosure.
- Learn to load code for CircuitPython, including libraries
- Install and run the code to test the DAMNED project.

Materials Needed

- Dozuki assembly tutorial which is on the Maker-E Dozuki site.
- Your completely assembled printed circuit board
- An LCD screen, acrylic diffuser, 3D printed button caps, FFC cable, assorted fasteners (distributed in class)
- A USB C cable for programming the ESP32
- Access to Thonny, the software used to program the ESP32

What you need to Know from Moodle Readings

- What basic hand and electronic tools are and how they are used for different tasks.

More Resources that may be Helpful

- Circuit Python Essentials guide on Adafruit

Relation to the DAMNED Project

So far we have worked on building and testing various pieces and parts of the project. This week is assembling all those parts together into a working DAMNED unit. This does not mean the project is done, since we have not really started on the software yet, but the hardware should be finished. After assembling the DAMNED project you'll download and install CircuitPython onto the ESP32 along with some needed libraries, then upload the code.py file that should show the project working as a whole.

What to Do and Know

This design assignment *should* be relatively quick and easy - provided everything goes well. However, in engineering often things don't work so well so please get started early. If you ran into some bottlenecks getting parts for the case made that is o.k., you should have some extra time this week. There are two parts to this assignment: assembling the unit from the various parts you have completed to date, then loading CircuitPython, libraries, and code. Each of these is described below.

1) Assembling the Device

After you have all of your parts designed, laser cut, and 3D printed, and you have a bag containing all of the other supplied materials and assembly hardware, it is time to assemble your device. Follow this Dozuki tutorial which will guide you through the assembly process. It should take under an hour to complete the assembly.

2) Programming - Running additional CircuitPython tests

Part of the Perusall reading for next week is on CircuitPython libraries and how files are organized. You will need a program called Thonny to modify and debug code. You will need to have the DAMNED plugged into your computer's USB port.

Update libraries. Follow this link and download the most up-to-date Bundle Version 10.x .zip file. After opening the .zip file you need to copy **ALL** of the following libraries to the /lib folder on your ESP32. If prompted, be sure to **overwrite** any existing files already on the device:

- adafruit_connection_manager.mpy
- adafruit_dotstar.mpy
- adafruit_requests.mpy
- adafruit_st7789.mpy
- adafruit_ntp.mpy
- adafruit_display_text (folder)

In addition to these standard CircuitPython libraries you will also need some additional libraries that are provided by the community of CircuitPython users. You can download these libraries in a .zip file from Moodle in the section where you downloaded this DA8 document. Open the .zip file and extract the contents into your /lib folder. You should have the following additional libraries:

- cp_mcp3x21.py
- wifiFunctions.py
- button_handler.py

Once you have these libraries updated then everything should work from a software perspective.

You should already have your settings.toml file configured with the correct WiFi settings from DA5. If not, please revisit DA5 and update this file accordingly. Remember that if this is your first time attempting to connect your ESP32 to the bucknell_iot network you must first register your device. Again,

see DA5 for instructions on how to complete this step.

Finally, find the DA8completeIntegrationTest.py file on Moodle. Download and save this code to your ESP32. Take a pin to pin jumper wire and connect it from the Analog_In port to a GND port on the back of your enclosure. Reference your IO label in order to properly make this connection. This should connect the input of your ADC to ground and provide an output of 0 V.

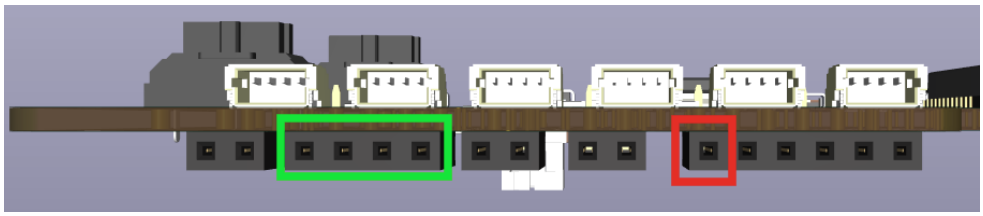


Figure 1: connect Analog_In (red) to any GND port (green)

With this connection made, execute the code in Thonny by opening DA8completeIntegrationTest.py and pressing F5 or clicking the green play button at the top. This will begin an automated test of all modules within your DAMNED. You should see feedback provided on the integrated LCD screen as test are initiated and completed. The order of the tests will be:

1. Basic LCD test of text output
2. Advanced LCD test with graphic display
3. ADC test
4. DotStar test
5. WiFi test
6. Button test

The final test of the buttons requires user input. Press each of the six buttons (up, down, left, right, A, B) at least once and confirm the correct button is acknowledged on the LCD. If any of the buttons are not acknowledged or not acknowledged correctly, be sure to note which button(s) is problematic and let Matt know.

The final test of the buttons is an infinite loop. Once you have tested all six buttons the tests are done and you can power down your DAMNED.

What to submit

You need to capture video of the six tests being performed on your DAMNED and submit the video to your Google Drive folder. Let the instructor know if you need to borrow a video capture device. The video needs to include the following elements to receive full credit for this assignment:

- Clear, well lit and consistent view of the entire front of your DAMNED enclosure. The LCD should

be visible and any output on the LCD should be readable in the video.
- Proper output for each of the six tests listed above
- The video should begin a few moments before the DA8completeIntegrationTest script is run
- Presses of all six buttons with an appropriate acknowledgement on the screen for each of the six button presses
- The video should end after all six buttons have been pressed and acknowledged on the screen

If your code doesn't execute properly, then you need to figure out where in the code an error is occurring or if there is a hardware error. One of the nice features of CircuitPython is that the serial monitor will tell you which line(s) in your code caused problems. Always begin by looking at the serial monitor, locate the line number that is problematic, and then inspect that line in your code. Often, errors will occur as a result of missing libraries.

The last element of DA8 is to complete two more sections of the course ePortfolio. Please visit Digication again and complete the following two sections:

- Technical Connections, Part 2 (DA8)
- Reflection, Part 1 (DA8)

Pay attention to what is being asked of you in this assignment. For "Technical Connections, Part 2 (DA8)" you are discussing a concept which is completely new to you that you identified during "Technical Connections, Part 1 (DA4)". You are not discussing one of the topics from ECEG 201. Be sure to look at the DA8 ePortfolio rubric on Moodle in order to receive full credit for the assignment.